Text-Grounded Interpretability for Mammography Risk Prediction via Vision–Language Alignment

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Introduction: State-of-the-art breast cancer risk models like MIRAI [1] demonstrate superior predictive performance but operate as black boxes, hindering clinical trust. While approaches such as AsymMIRAI [2] provide spatial attribution via heatmaps, they fail to offer text-interpretability. Clinicians reason through report descriptors, not heatmaps alone. To bridge this gap, we introduce Mammo-FM [3], a mammography specific foundation model (FM) that transforms these black-box risk predictors into textually interpretable models by leveraging vision-language alignment.

Objectives: 1. Build a large mammography-specific FM using multi-institution pretraining. The cohort includes **96,567 Mayo patients, 13,829 UPMC patients, 9,900 BU patients, and 20,381 EMBED patients** (total **140, 677 patients; 821,326 mammograms**). 2. Transform MIRAI and AsymMIRAI into **textually interpretable** risk predictors by leveraging Mammo-FM's aligned visual and textual features.

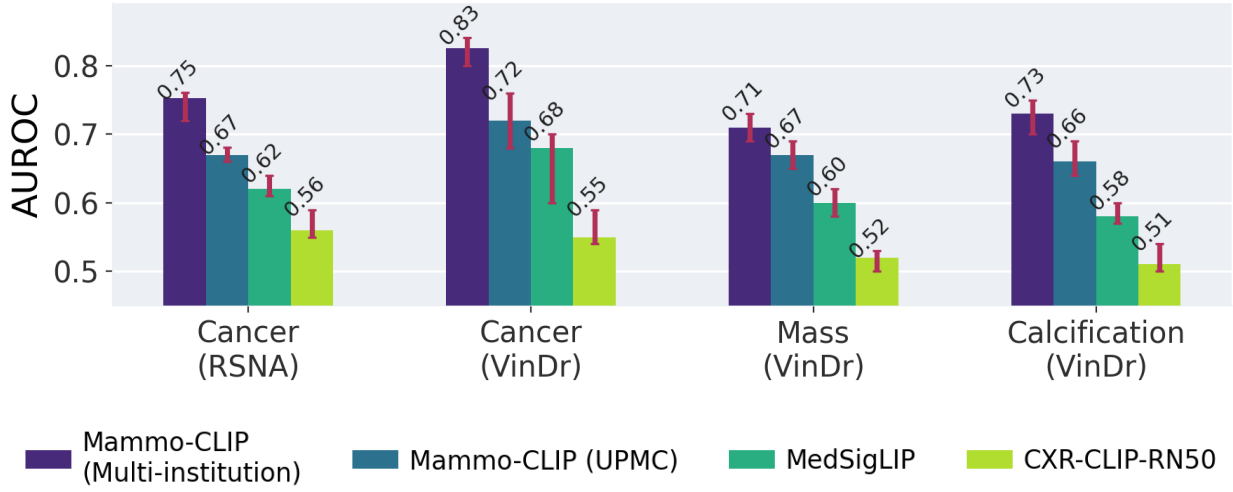


Figure 1: Zero-shot evaluation of breast cancer and finding classification performance on out-of-distribution – VinDr (classifying cancer, mass and calcification) and RSNA (classifying cancer) - datasets.

Method: Mammo-FM employs a dual-encoder architecture optimizing a multiview contrastive loss. The vision encoder utilizes EfficientNet-B5 initialized on ImageNet, while

Modern-BERT serves as the text encoder. We validate diagnostic robustness with **training-free zero-shot** classification on public OOD datasets: **RSNA** (classifying cancer; 11,913 patients; 54,706 mammograms) and **VinDr-Mammo** (classifying cancer, mass, calcification; 5,000 patients; 20,486 mammo grams). We benchmark against MedSigLIP [5], a UPMC-only Mammo-FM variant and CXR-CLIP [5] chest X-ray FM baselines. For risk interpretability, we integrate Mammo-FM's report-align

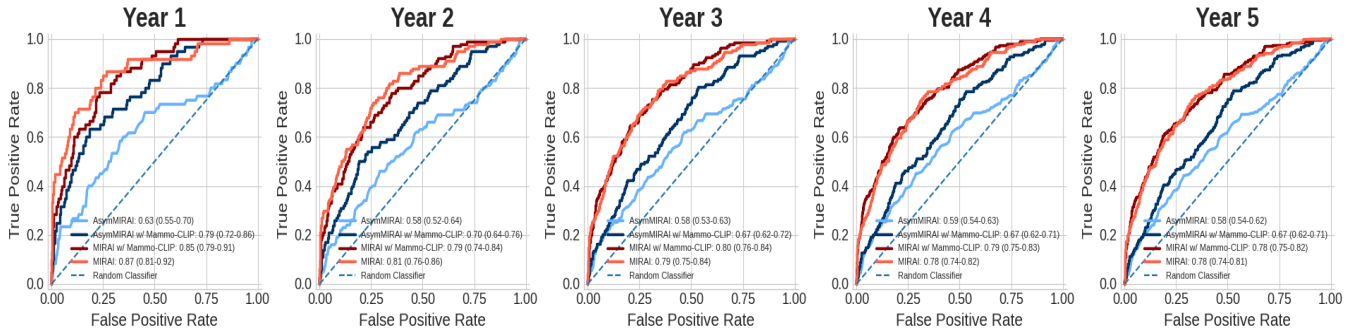
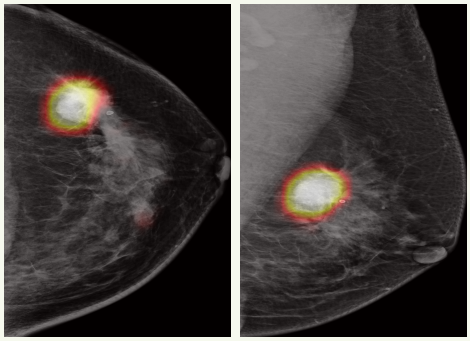


Figure 2: Comparison of AUROC for 1–5-year risk prediction MIRAI w/ Mammo-FM, AsymMIRAI w/ Mammo-FM, and their baselines using BU dataset, showing that Mammo-FM-based encoders retain the performance of the original models while maintaining interpretability.

ed frozen vision encoder into MIRAI and AsymMIRAI, yielding MIRAI w/ Mammo-FM and AsymMIRAI w/ Mammo-FM, and we train both via knowledge distillation from MIRAI risk outputs. We then introduce a mechanistic interpretability layer with a Matryoshka Sparse Autoencoder (SAE), trained on Mammo-FM's spatial patch features to preserve lesion localization. We train the SAE on ~1.2M patch embeddings from the pretraining cohort, extracted on a 48×29 spatial grid. The SAE expands 2048-d patch features into a 16,384-d sparse latent space and enforces hierarchical sparsity through multi-scale reconstruction. This design yields neuron-level concept units that capture distinct mammographic patterns and retain discriminative fidelity. We quantify causal neuron influence with Monte Carlo Shapley estimation and rank risk-increasing neurons by contribution magnitude. We localize evidence with neuron activation heatmaps and ground each neuron in language via causal ablation, selecting the radiology-report sentence with the largest image–text similarity drop. We evaluate 1–5 year risk prediction in-distribution on BU screening exams (13,857 exams; 55,428 images). **Results:** Fig. 1 illustrates that Mammo-FM achieves strong diagnostic transfer under zero-shot evaluation and delivers **13–38%** higher AUROC than MedSigLIP across tasks. For cancer classification, Mammo-FM reaches AUROC **0.75** (95% CI: 0.72–0.76) on RSNA and AUROC **0.83** (95% CI: 0.80–0.84) on VinDr. Fig. 2 demonstrates that Mammo-FM-based risk predictors preserves MIRAI-level risk performanc. MIRAI w/ Mammo-FM achieves AUROC **0.85** (95% CI: 0.79–0.91) versus **0.87** (95% CI: 0.81–0.92) for MIRAI, which confirms that report-aligned features retain predictive

strength while enabling



Findings: There are no prior studies for comparison. Both breasts show scattered fibroglandular densities. No suspicious dominant mass lesions, clustered microcalcifications or areas of unexplained architectural distortion are seen in the right breast. There is a mass in the lateral left breast that needs additional imaging. **Impression: BI-RADS: 0. Mass in the left breast needs additional imaging.**

Figure 3: Qualitative visualization of MIRAI w/ Mammo-FM risk interpretation for a high-risk 1-year case (MIRAI risk = 0.74) in the BU dataset shows dual interpretability: image-based saliency (left) highlights a neuron selective for mass features, while the corresponding report sentence (right)yields the highest similarity drop for risk, indicating language-based interpretability.

interpretability. In a high-risk 1-year case (risk 0.74), Shapley heatmaps localize the top neuron to a left-breast mass, and neuron ablation selects the paired report sentence: “Mass in the left breast needs additional imaging.” (Fig. 3)

Key Findings: We demonstrate that vision-language alignment in Mammo-FM transforms black-box risk predictors interpretable by grounding report-aligned visual representations in sentence-level clinical language while preserving predictive robustness. **Impact on Screening Strategies**: Mammo-FM grounds risk in clinical language, strengthening trust, enabling audit, and supporting confident screening-interval tailoring for clinical deployment.

References: [1] Yala et al.Toward robust mammography-based models for breast cancer risk. Science Translational Medicine 13.578 (2021). [2] Donnelly et al. AsymMirai: Interpretable Mammography-based Deep Learning Model for 1–5-year Breast Cancer Risk Prediction. Radiology: Artificial Intellignece2024. [3] Ghosh et al. Mammo-FM: Breast-specific foundational model for Integrated Mammographic Diagnosis, Prognosis, and Reporting. ArXiv 2025. [4] Sellergren et al. MedGemma Technical Report. ArXiv 2025. [5] You et al. CXR-CLIP: Toward Large Scale Chest X-ray Language-Image Pre-training. MICCAI 2023